

Proof. **Step1. Degenerate cases.** If $S_{b^2} = 0$ then each $b_i = 0$; consequently $S_{ab} = 0$ and

$$(S_{ab})^2 = 0 \leq S_{a^2} S_{b^2} = 0.$$

The same argument works when $S_{a^2} = 0$. Hence the inequality is trivial in these cases. In what follows we assume

$$S_{a^2} > 0, \quad S_{b^2} > 0.$$

Step2. An auxiliary quadratic function. Define

$$Q(t) = \sum_{i=1}^n (a_i - t b_i)^2, \quad t \in \mathbb{R}.$$

Expanding each term,

$$(a_i - t b_i)^2 = a_i^2 - 2t a_i b_i + t^2 b_i^2,$$

and summing over i gives

$$\begin{aligned} Q(t) &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2 \\ &= S_{a^2} - 2t S_{ab} + t^2 S_{b^2}. \end{aligned} \tag{2.1}$$

Since each summand $(a_i - t b_i)^2$ is non-negative,

$$Q(t) \geq 0 \quad \text{for every } t \in \mathbb{R}. \tag{2.2}$$

Step3. Discriminant argument. Equation (2.1) shows that $Q(t)$ is a quadratic polynomial in t with

$$A = S_{b^2}, \quad B = -2S_{ab}, \quad C = S_{a^2}.$$

Because $A > 0$ and $Q(t) \geq 0$ for all real t , the discriminant must be non-positive:

$$B^2 - 4AC \leq 0.$$

Explicitly,

$$(-2S_{ab})^2 - 4 S_{b^2} S_{a^2} = 4(S_{ab}^2 - S_{a^2} S_{b^2}) \leq 0.$$

Dividing by the positive number 4 yields

$$S_{ab}^2 \leq S_{a^2} S_{b^2}. \tag{3.1}$$

This is precisely the Cauchy–Schwarz inequality

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

Step4. Equality condition. Equality in (3.1) occurs exactly when the discriminant is zero, i.e. $B^2 - 4AC = 0$. In this case the quadratic $Q(t)$ has the unique real root

$$t_0 = -\frac{B}{2A} = \frac{S_{ab}}{S_{b^2}}.$$

Evaluating Q at t_0 using (2.1),

$$Q(t_0) = \frac{S_{a^2}S_{b^2} - S_{ab}^2}{S_{b^2}} = 0.$$

Since each term $(a_i - t_0 b_i)^2$ is non-negative, $Q(t_0) = 0$ forces

$$a_i = t_0 b_i \quad \text{for all } i = 1, \dots, n.$$

Thus equality holds iff the vectors $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ are linearly dependent, i.e. one is a scalar multiple of the other. The degenerate cases $S_{a^2} = 0$ or $S_{b^2} = 0$ fit this description because the zero vector is a multiple of every vector.

Step5. Conclusion. For all real sequences $\{a_i\}, \{b_i\}$ we have

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

with equality if and only if the two vectors are linearly dependent (including the trivial zero-vector cases). $\square$